

LESSON 1.5

SOLVING ONE-STEP EQUATIONS



LESSON INTRODUCTION

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

LEARNING TARGETS

- I can solve one step equations involving addition, subtraction, multiplication, and division using models and inverse operations.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

ESSENTIAL QUESTIONS

- How can I solve one step equations using inverse operations?

LESSON NARRATIVE

This lesson reviews solving one-step algebraic equations using inverse operations. Students complete an Exploration using algebra tiles to show a visual understanding of solving equations. Lesson 4 serves as a bridge between operations with integers and real-world applications in Lesson 6.

COMPONENT SUMMARY

1.5.1 WARM-UP (~ 5 MIN.)

Practice with ratios

1.5.3 COLLABORATION (5 MIN.)

Collaborate on equivalent inverse operations

1.5.5 COLLABORATION (5 MIN.)

Collaborate on solving one-step equations using inverse operations

1.5.2 GUIDED PRACTICE (20 MIN.)

Review of solving one-step equations using inverse operations

1.5.4 TRY IT (10 MIN.)

Practice using inverse operations to solve one-step equations

1.5.6 WRAP-UP (~5 MIN.)

Self-reflection on the lesson

RESOURCES

GLOSSARY

Key Terms:

- Inverse operation

Additional Terms:

- Equation

MATERIALS

- (Optional) Algebra tiles

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may get stuck understanding what operation is shown in division equations (i.e., $\frac{x}{5} = 3$) because they will see it as a fraction and not see the division.
- Students may have a hard time determining the inverse operation when the variable is on the right side of the equation.
- Students may struggle with operations with integers.

THINGS TO CONSIDER

- Consider using the information from the self-reflection in Lesson 4 to emphasize the operations with integers that students struggled with the most.
- Consider providing sentence frames to support students to produce statements with justification during conversation.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 1.5.3 Collaboration, provide students with the opportunity to independently read, process, and answer question 1b before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of the information, first for independent learners, then pairing for verbal processing to give an opportunity for more confident, precise responses when students share out.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In this lesson, students are provided the opportunity to collaborate with their partners. In the first collaboration partners work together to consider two different students' strategies in solving a real-world equation. Students discuss and compare the efficiency of the work of two students. In the second collaboration partners work together to verify each other's answers. Students recognize errors and suggest to their partner how to correctly solve the problem.

DIFFERENTIATE INSTRUCTION

The On-Ramp to 7th Grade Accelerated Math video and practice tool offers a variety of previously learned skills instruction and practice to support all students. Consider having students preview this lesson by watching the On-Ramp video or have the video available to watch for students to opt-in throughout the lesson.

- Solving Equations (Addition)
- Solving Equations (Multiplication)

1.5.1 WARM-UP: BRAIN CAPACITY

ENRICHMENT

What are different ways we can represent this ratio?



1.5.1 Warm-Up: Brain Capacity

A human brain can store about 3,500 trillion bytes of data, while a cat's brain can store about 98 trillion bytes of data.

1. What is the brain storage capacity of human to cat?
 $3,500 : 98$
Human:cat
2. About how many brains of cats would it take to store the same amount of data as the human brain?
There are about 35 cats needed to store the same amount of data as the human brain.



1.5.2 Guided Practice: Solving Equations

1. Four algebraic equations are represented using algebra tiles.

Model A

Model B

Model C

Model D

a. Match the algebra tile model to the equations.

Equation	Model
$4x = 8$	<i>D</i>
$\frac{x}{4} = 8$	<i>B</i>
$8 = x + 4$	<i>A</i>
$x - 4 = 8$	<i>C</i>

b. Use the algebra tile models to determine the value of x in each equation.

Equation	Value of x
$4x = 8$	$x = 2$
$\frac{x}{4} = 8$	$x = 32$
$8 = x + 4$	$x = 4$
$x - 4 = 8$	$x = 12$

1.5.2 GUIDED PRACTICE: SOLVING EQUATIONS

DIFFERENTIATE INSTRUCTION

Consider having students model the equations with actual algebra tiles on their desk.

ENRICHMENT

Students may pause when determining how to interpret the algebra tiles. Consider prompting with the following questions:

- What is the difference between the red unit tiles and blue unit tiles?
- What does it mean to have more than one x on one side of the equation?
- How is division represented with algebra tiles?

DIFFERENTIATE INSTRUCTION

Prompt students to continue using their algebra tiles and manipulating them to determine the value of x in each equation.

1.5.2 GUIDED PRACTICE: SOLVING EQUATIONS (CONTINUED)

TEACHER MOVES

Prompt students for a verbal description of $\frac{x}{4} = 8$ to identify any potential gaps for $\frac{b}{-4} = 13$ in question 4.

ENRICHMENT

What would be the inverse of adding a negative number?

DIFFERENTIATE INSTRUCTION

Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems and provide ample time for students to grapple with academic language:

_____ is the inverse operation of _____, so to solve this equation we have to _____.

TEACHER MOVES

Students may struggle to recognize that $\frac{b}{-4}$ represents division. If they do, consider asking students how they know that $\frac{1}{2} = 0.5$. Show the long division to help a student connect that $\frac{b}{-4}$ is the same as $b \div -4$.

ENRICHMENT

How can you verify that the solution is correct?

- c. Choose one of the equations and explain how you determined the value of x from the model.

Answers will vary.

For $x - 4 = 8$, I counted to see how many ones were needed to balance the equation.

When finding the solution to an equation, **inverse operations** are used to balance an equation.



An **inverse operation** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction (-3)	Addition $+3$
Addition ($+6$)	Subtraction -6
Multiplication ($\times 1.5$)	Division $\div 1.5$
Division ($\div 10$)	Multiplication $\times 10$

3. Use the equation $21 = 7h$ to complete the following.
- What mathematical operation is represented by $7h$?
Multiplication
 - What is the inverse operation?
Division
 - Perform the inverse operation on both sides of the equation. Show your work.

$$\begin{array}{r} 21 = 7h \\ \hline 7 \quad 7 \end{array}$$

- What is the value of h ?
 $h = 3$
- Discuss with your partner how to check whether or not your solution is correct. Then, check your solution.
Answers may vary. To check if my solution is correct, I can plug in the value for h in the equation.
 $21 = 7h$
 $21 = 7(3)$
 $21 = 21$

4. Use the equation $\frac{b}{-4} = 13$ to complete the following:
- What mathematical operation is represented by $\frac{b}{-4}$?
Division
 - What is the inverse operation?
Multiplication
 - Perform the inverse operation on both sides of the equation. Show your work.

$$-4 \times \frac{b}{-4} = 13 \times -4$$

- What is the value of b ?
 $b = -52$
- Verify your solution is correct.
 $\frac{-52}{-4} = 13$
 $13 = 13$



1.5.3 Collaboration: Choose a Strategy

1. An elevator starts in the parking garage under a building at level -3 , which is 3 levels underground. The elevator travels to the sixth floor of the building. This situation can be modeled by the equation $-3 + x = 6$, where x represents the number of floors the elevator travels.

Tyler says that since subtraction is the inverse operation of addition, you should use subtraction to solve. Eva says that you should add 3 to both sides of the equation to solve it.

- a. Complete the table by showing the result of each student's strategy.

Tyler's Strategy	Eva's Strategy
$(-3) + x = 6$	$(-3) + x = 6$
$-(-3) \quad -(-3)$	$+3 \quad +3$
$x = \underline{9}$	$x = \underline{9}$

- b. With your partner, determine whose strategy is correct. Explain how you know.
Answers will vary.
Both strategies are correct because adding 3 and subtracting -3 has the same result.

1.5.3 COLLABORATION: CHOOSE A STRATEGY

DIFFERENTIATE INSTRUCTION

Consider prompting students to create a visual representation of this situation for an alternative entryway into solving this problem.

TEACHER MOVES

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions in order to identify which students surface those misconceptions and which students help clarify.

TEACHER MOVES

Provide students with the opportunity to independently read, process, and answer question 1b before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of the information, first for independent learners, then pairing for verbal processing in order to give an opportunity for more confident, precise responses when students share out.

1.5.4 TRY IT: SOLVING EQUATIONS

TEACHER MOVES

For the equation $-36 = \frac{m}{-6}$ see the common misconception from above and consider prompting students to rewrite the equation if they are struggling to determine the inverse operation.

TEACHER MOVES

Monitor student work as they work on their own to solve the equations. Their approach might illuminate potential gaps. For example, Students may get 6 for the third equation, indicating a gap in understanding the relationship with division when represented as a fraction.



1.5.4 Try It: Solving Equations

- Each equation shown in the table can be solved in one step. Complete the table by describing how to solve each equation, showing your work, and providing the solution. The first row is started for you.

Equation	Description	Work	Solution	Verify
$6x = 18$	divide both sides of the equation by 6	$\frac{6x}{6} = \frac{18}{6}$ $x = 3$	3	$6(3) = 18$ $18 = 18$
$h - 5 = -7$	add 5 to both sides of the equation	$\begin{array}{r} h - 5 = -7 \\ +5 \quad +5 \\ \hline h = -2 \end{array}$	-2	$-2 - 5 = -7$ $-7 = -7$
$-36 = \frac{m}{-6}$	multiply both sides of the equation by -6	$(-6) \times -36 = \frac{m}{-6} \times (-6)$	216	$-36 = \frac{216}{-6}$ $-36 = -36$
$32 = 41 + a$	subtract 41 from both sides of the equation	$\begin{array}{r} 32 = 41 + a \\ -41 \quad -41 \\ \hline -9 = a \end{array}$	-9	$32 = 41 + (-9)$ $32 = 32$
$-45 = 9y$	divide both sides of the equation by 9	$\frac{-45}{9} = \frac{9y}{9}$ $-5 = y$	-5	$-45 = 9(-5)$ $-45 = -45$

- Compare your work with your partner and resolve any differences. Verify your solutions using substitution.

1.5.5 COLLABORATION: SOLVING EQUATIONS

TEACHER MOVES

Prompt students to utilize substitution to verify their answers.

ENRICHMENT

What strategies or methods did you use when solving the equations? Which problems did you and your partner have different answers for? What mistake was made?

QUESTIONING

For students who finish early, consider the On-Ramp to 7th Grade Accelerated Math learning tool: Solving Equations as additional practice for students.



1.5.5 Collaboration: Solving Equations

With your partner, determine who will be Partner A and who will be Partner B.

- Solve the equations from your designated column.

Partner A	Partner B
$72 = -9x$ $\frac{72}{-9} = \frac{-9x}{-9}$ $x = -8$	$x - 4 = -12$ $x - 4 + 4 = -12 + 4$ $x = -8$
$469 + z = 459$ $469 - 469 + z = 459 - 469$ $z = -10$	$\frac{z}{5} = -2$ $\frac{z}{5} \cdot (5) = -2 \cdot (5)$ $z = -10$
$3 = \frac{w}{-6}$ $(-6) \cdot 3 = \frac{w}{-6} \cdot (-6)$ $w = -18$	$-2w = 36$ $\frac{-2w}{-2} = \frac{36}{-2}$ $w = -18$
$-6 = x - 20$ $-6 + 20 = x - 20 + 20$ $x = 14$	$-8 = -22 + x$ $-8 + 22 = -22 + 22 + x$ $x = 14$

- The equations in each row have the same solutions. Check your work with your partner, making revisions to your work if necessary.



1.5.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is related to each target.

- ☐ I can solve one-step equations involving addition and subtraction.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



- ☐ I can solve one-step equations involving multiplication and division.

I need help. I'm getting there, but I understand this
I can't get started. I need more practice. and I feel confident.



Answers will vary.

1.5.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider using information gathered throughout the lesson as well as in this Wrap-Up to inform instruction in Lesson 6, particularly for making groups in Collaboration activities.